

LESSON PURPOSE

One class period

Students inspect six purpose-made samples, identify the joining method, classify the joint, and justify when each method is appropriate. The goal is not memorizing hardware names; the goal is selecting a connection that fits the interface requirements.

SAMPLE KIT

- Station A: Bolt + nut + washer**
Two coupons with aligned through holes. Include bolt, nut, and washer. Students should be able to assemble and disassemble the joint.
- Station B: Screw into threads**
Top clearance hole plus lower printed/tapped/threaded feature, insert, or captured nut. Shows that a screw can fasten into one part.
- Station C: Riveted overlap joint**
Two thin coupons joined by a pop rivet. Keep this station permanently assembled.
- Station D: Adhesive lap joint**
Two overlapping coupons bonded with hot glue, CA glue, epoxy, or another classroom-safe adhesive. Keep assembled.
- Station E: Press/friction fit**
A pin and block with three holes: loose, snug, and tight. This should visibly connect back to tolerance and fit.
- Station F: Tab/slot + snap feature**
Two printed parts where one feature locates the parts and a flexible snap feature retains them.

RECOMMENDED CLASS FLOW

- 0-10 min**
Direct instruction: joint categories, removable/semi-permanent/permanent, and decision factors.
- 10-35 min**
Station rotations. Keep groups moving. The worksheet is designed for concise observations, not essays.
- 35-50 min**
Engineering selection scenarios and deep comparison. Push students to justify with at least three factors.
- Final 5 min**
Whole-class debrief: which connection is easiest to service, most manufacturing-dependent, and most likely to fail if dimensions are poor?



EXPECTED IDENTIFICATION

- A. Bolt, nut, washer - removable mechanical fastener; strong clamping; serviceable; requires hardware/access.
- B. Screw connection - removable mechanical fastener; fewer loose parts; thread engagement matters.
- C. Rivet - permanent or semi-permanent mechanical fastener; fast; poor serviceability.
- D. Adhesive lap joint - permanent bonded joint; no hardware; depends on surface preparation and bond area.
- E. Press/friction fit - fit-based joint; dimensional control determines function.
- F. Tab/slot + snap feature - integrated/form-based joint; fast assembly; flexible features can wear or break.

SCENARIO GUIDANCE

Avionics access panel

Best answers usually select screws or bolts because the panel must be removable and vibration-resistant.

Permanent thin-sheet bracket

Rivets or adhesive can be defended. Rivets are strong, fast, and repeatable; adhesive can reduce hardware but depends more on process control.

Printed sensor housing

Snap fit/tab-and-slot or screws can be defended. Snap features minimize hardware and speed assembly; screws improve serviceability and durability.

LOOK-FORS WHILE GRADING

- Student identifies both the method and the interface function.
- Student distinguishes removable, semi-permanent, and permanent connections.
- Student uses multiple decision factors, not a single phrase like stronger or easier.
- Student recognizes that press fits and snap fits depend on manufacturing accuracy.
- Student can compare an existing method with a reasonable alternative.